

CYBERSHIELD - CYBERSECURITY MONITORING TOOLKIT

1. Password Strength Checker –

Input :-

```
1 import re
2 from datetime import datetime
3
4
5 def password_checker():
6     password = input("Enter Password : ")
7
8     score = 0
9
10
11     if len(password) >= 8:
12         score += 1
13
14     if re.search(r"[A-Z]", password):
15         score += 1
16
17     if re.search(r"[a-z]", password):
18         score += 1
19
20     if re.search(r"\d", password):
21         score += 1
22
23     if re.search(r"[!@#$%^&*(),.?\"':{}|<>]", password):
24         score += 1
25
26     if score <= 2:
27         strength = "WEAK"
28
29     elif score <= 4:
30         strength = "MODERATE"
31
32     else:
33         strength = "STRONG"
34
35     print("\nPassword Strength :", strength)
36
37     with open("scan_history.txt", "a") as file:
38         file.write(f"\n{datetime.now()}\n")
39         file.write(f"Password Strength : {strength}\n")
```

Output :-

```
=====
CYBERSHIELD TOOLKIT
=====
1. Password Strength Checker
2. Port Scanner
3. Security Log Scanner
4. Website Availability Checker
5. View Scan History
6. Exit

Enter Choice : 1
Enter Password : Saksham@69

Password Strength : STRONG
```

2. Log File Scanner

Code :-

```
1 from datetime import datetime
2 import os
3
4 def log_scanner():
5
6     base_dir = os.path.dirname(os.path.abspath(__file__))
7
8     (variable) history_file: str, "security_log.txt")
9     history_file = os.path.join(base_dir, "scan_history.txt")
10
11     if not os.path.exists(log_file):
12         print("\nsecurity_log.txt not found!")
13         print("Expected location:")
14         print(log_file)
15         return
16
17     keywords = ["ERROR", "FAILED LOGIN", "WARNING"]
18
19     with open(log_file, "r") as file:
20         data = file.read().upper()
21
22     print("\nSecurity Log Scan Report")
23     print("-----")
24
25     history = ""
26
27     for word in keywords:
28         count = data.count(word)
29         print(f"{word} : {count}")
30         history += f"{word} : {count}\n"
31
32     with open(history_file, "a") as file:
33         file.write(f"\n{datetime.now()}\n")
34         file.write("Log Scanner\n")
35         file.write(history)
36
```

Output :-

```
=====
CYBERSHIELD TOOLKIT
=====
1. Password Strength Checker
2. Port Scanner
3. Security Log Scanner
4. Website Availability Checker
5. View Scan History
6. Exit

Enter Choice : 3

Security Log Scan Report
-----
ERROR : 3
FAILED LOGIN : 3
WARNING : 2

Press Enter... █
```

3. Port Scanner

Code :-

```
import socket
from datetime import datetime
def port_scanner():

    ip = input("Enter IP Address : ")

    ports = {
        21: "FTP",
        22: "SSH",
        80: "HTTP",
        443: "HTTPS"
    }

    print("\nScanning...\n")

    history = ""

    for port, service in ports.items():

        sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        sock.settimeout(1)

        result = sock.connect_ex((ip, port))

        if result == 0:
            status = "Open"
        else:
            status = "Closed"

        print(f"{port} ({service}) : {status}")

        history += f"{port} {service} : {status}\n"

        sock.close()

    with open("scan_history.txt", "a") as file:
        file.write(f"\n{datetime.now()}\n")
        file.write(f"Port Scan : {ip}\n")
        file.write(history)
```

Output :-

```
=====
CYBERSHIELD TOOLKIT
=====
1. Password Strength Checker
2. Port Scanner
3. Security Log Scanner
4. Website Availability Checker
5. View Scan History
6. Exit

Enter Choice : 2
Enter IP Address : 192.169.0.2

Scanning...

21 (FTP) : Closed
22 (SSH) : Closed
80 (HTTP) : Closed
443 (HTTPS) : Closed

Press Enter...█
```

4. Website Availability Checker

Code :-

```
1 ~ import requests
2   from datetime import datetime
3
4
5 ~ def website_checker():
6
7     website = input("Enter Website URL : ")
8
9 ~     try:
10
11         response = requests.get(website, timeout=5)
12
13         print("\nWebsite is Online")
14
15         print("Status Code :", response.status_code)
16
17         status = "Online"
18
19 ~     except:
20
21         print("\nWebsite is Offline")
22
23         status = "Offline"
24
25 ~     with open("scan_history.txt", "a") as file:
26
27         file.write(f"\n{datetime.now()}\n")
28
29         file.write(f"Website : {website}\n")
30
31         file.write(f"Status : {status}\n")
```

Output :-

```
=====
CYBERSHIELD TOOLKIT
=====
1. Password Strength Checker
2. Port Scanner
3. Security Log Scanner
4. Website Availability Checker
5. View Scan History
6. Exit

Enter Choice : 4
Enter Website URL : https://www.instagram.com

Website is Online
Status Code : 200
```

5. Scan History Viewer

Code :-

```
1  import os
2
3
4  def view_history():
5
6      if not os.path.exists("scan_history.txt"):
7
8          print("No History Found")
9
10         return
11
12     with open("scan_history.txt", "r") as file:
13
14         print("\n")
15
16         print(file.read())
```

EXIT OUTPUT :-

```
=====
CYBERSHIELD TOOLKIT
=====
1. Password Strength Checker
2. Port Scanner
3. Security Log Scanner
4. Website Availability Checker
5. View Scan History
6. Exit

Enter Choice : 6

Thank You for using CyberShield
PS C:\Users\Saksham\OneDrive\Desktop\Cybersecurity Journey> █
```

Output:-

```
=====
CYBERSHIELD TOOLKIT
=====
1. Password Strength Checker
2. Port Scanner
3. Security Log Scanner
4. Website Availability Checker
5. View Scan History
6. Exit

Enter Choice : 5

2026-08-07 14:00:07.859007
Password Strength : STRONG

2026-08-07 Open file in editor (ctrl + click)
Website : C:\Users\Saksham\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Sak
Status : Offline

2026-08-07 14:39:52.975372
Port Scan : 192.169.0.2
21 FTP : Closed
22 SSH : Closed
80 HTTP : Closed
443 HTTPS : Closed

2026-08-07 14:42:43.896913
Website : https://www.instagram.com
Status : Online

Press Enter...|
```